20/4/2026

$$D[f(x)] = F(x)$$

$$\overset{?}{=} \int F(x)$$

$$\boxed{\begin{aligned} I &= \int a \sin x \, dx \\ \frac{dI}{da} &= \frac{d}{da}\int a \sin x \end{aligned}}$$

Integration & differentiation are <u>inverse operations</u>

$$f'(x) = F(x)$$

ie $\frac{d}{dx} f(x) = F(x)$

ie $df(x) = F(x)\,dx$.

$$\Rightarrow \int df(x) = \int F(x)\,dx$$

$$\downarrow$$

$$f(x)$$

I

① dI

② $dI = I$

DUIS: Leibnitz rule of Differentiation under the integral sign

Suppose

$$I(\alpha) = \int_a^b \underline{f(x,\alpha)}\, dx$$

Where α is a parameter, and $f(x,\alpha)$ is differentiable in x (w.r.t) & α in $[a,b]$ then

$$\frac{dI}{d\alpha} = \frac{d}{d\alpha}\int_a^b \underline{f(x,\alpha)}\, dx = \int_a^b \frac{\partial}{\partial\alpha} f(x,\alpha)\, dx \quad \hookrightarrow (*)$$

$(*)$ is the Leibnitz rule of DUIS for one parameter and constant limits.

Note: DUIS method is used for evaluating integrals which may not be integrated by the usual methods.

(1) $f(x) \leftarrow I = \int \cos x\, dx = \sin x + c$

(2) $I = \int_0^{\pi/2} \cos x\, dx = \sin x \Big]_0^{\pi/2}$
$= 1$

(3) $I = \int_0^{\pi/2} \alpha \cos x\, dx = \alpha \sin x \Big]_0^{\pi/2}$
$= \underline{\underline{\alpha}}$

Egs

1. Prove that $\int_0^1 \frac{x^\alpha - 1}{\log x}\,dx = \log(1+\alpha)$, $\alpha \geq 0$

Sol^n: Let

$$I(\alpha) = \int_0^1 \frac{x^\alpha - 1}{\log x}\,dx \quad \text{— (1)}$$

By DUIS rule

$$\frac{dI(\alpha)}{d\alpha} = \int_0^1 \frac{\partial}{\partial\alpha}\left[\frac{x^\alpha - 1}{\log x}\right] dx$$

$$\left[\frac{d}{dx}(a^x) = a^x \log a \quad \therefore \frac{d}{da}(x^a) = x^a \log x\right]$$

$$= \int_0^1 \frac{1}{\cancel{\log x}}\left[x^\alpha \cancel{\log x} - 0\right] dx$$

$$\Rightarrow \frac{dI}{d\alpha} = \int_0^1 x^\alpha\,dx$$

$$= \left.\frac{x^{\alpha+1}}{\alpha+1}\right]_0^1$$

$$= \frac{1}{\alpha+1}\left[1 - 0\right]$$

$$\Rightarrow \frac{dI}{d\alpha} = \frac{1}{\alpha+1}$$

$$\Rightarrow \int dI = \int \frac{1}{\alpha+1}\,d\alpha$$

$$\Rightarrow I = \log(\alpha+1) + C \quad \text{— (2)}$$

putting $\alpha = 0$ in (1) we get

$$I(0) = \int_0^1 \frac{x^0 - 1}{\log x}\,dx = 0$$

$\alpha = 0$ in (2) gives:

$$I(0) = \log(0+1) + C$$

ie $0 = \log 1 + C$

using (1)

ie $0 = 0 + C$

$\Rightarrow \boxed{C = 0}$

$$\therefore \int_0^1 \frac{x^\alpha - 1}{\log x}\,dx = I(\alpha) = \log(\alpha+1)$$

2. P.T $\int_0^1 \frac{x^\alpha - x^\beta}{\log x}\,dx = \log\left(\frac{1+\alpha}{1+\beta}\right)$

<u>Soln</u>: Let

$$I_1 = \int_0^1 \frac{x^\alpha - 1}{\log x}\,dx$$

$$\& \; I_2 = \int_0^1 \frac{x^\beta - 1}{\log x}\,dx$$

Then $\int_0^1 \frac{x^\alpha - x^\beta}{\log x}\,dx = \int_0^1 \frac{(x^\alpha - 1) - (x^\beta - 1)}{\log x}\,dx$

$$\Rightarrow \int_0^1 \frac{x^\alpha - x^\beta}{\log x}\,dx = I_1 - I_2$$

$$= \log(1+\alpha) - \log(1+\beta)$$

(Refer eg 1)

$$= \log\left(\frac{1+\alpha}{1+\beta}\right)$$

3. P.T $\int_0^\infty \left(\frac{1-e^{-ax}}{x}\right) e^{-x}\, dx = \log(a+1)$

Sol^n: Let

$$I(a) = \int_0^\infty \left(\frac{1-e^{-ax}}{x}\right) e^{-x}\, dx \quad \text{— (1)}$$

By the rule of DUIS,

$$\frac{dI}{da} = \int_0^\infty \frac{\partial}{\partial a}\left(\frac{1-e^{-ax}}{x}\right) e^{-x}\, dx$$

ie $$\frac{dI}{da} = \int_0^\infty \frac{e^{-x}}{x}\left[0-(-xe^{-ax})\right] dx$$

$$\Rightarrow \frac{dI}{da} = \int_0^\infty \frac{e^{-x}}{\cancel{x}} \cdot \cancel{x} e^{-ax}\, dx$$

$$\Rightarrow \frac{dI}{da} = \int_0^\infty e^{-x(1+a)}\, dx$$

$$= \left.\frac{e^{-x(1+a)}}{-(1+a)}\right]_0^\infty$$

$$\Rightarrow \frac{dI}{da} = -\frac{1}{(1+a)}\left[0-1\right]$$

$$\Rightarrow \frac{dI}{da} = \frac{1}{1+a}$$

$$\Rightarrow \int dI = \int \frac{1}{1+a}\, da$$

$$\Rightarrow I = \log(1+a) + c \quad \text{— (2)}$$

Put $a=0$ in (1),

$I(0) = 0$

Using this in (2),

$I(0) = 0 = \log(1+0) + c$

ie $0 = \log 1 + c$

$\Rightarrow 0 = 0 + c \Rightarrow \boxed{c=0}$

$\therefore \underline{I = \log(1+a)}$